

Let T be a field, $a \in T$ and $A \in T^{n \times n}$ is a square matrix, such that the sum of all elements in every column is equal to a . Prove that a is an eigenvalue of A

Using $\det A = \det A^T$ notice that

$$\det(A - \lambda I) = \det(A - \lambda I)^T = \det(A^T - \lambda I^T) = \det(A^T - \lambda I)$$

or in other words: λ is an eigenvalue of A if and only if it's an eigenvalue of A^T . However since the sum of all elements in a column of A is a , the same is true for rows in A^T . Let $A^T = (\tilde{a}_1 | \cdots | \tilde{a}_n)$, then from the previous observation

$$\tilde{a}_1 + \cdots + \tilde{a}_n = \sum_{k=1}^n \tilde{a}_k = \begin{pmatrix} a \\ \vdots \\ a \end{pmatrix}$$

Which can be rewritten as

$$A^T \cdot \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ \vdots \\ a \end{pmatrix} \quad \text{or} \quad A^T \vec{v} = a \vec{v} \quad ; \quad \vec{v} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \neq \vec{0}$$

Therefore a is an eigenvalue of A^T and also A .

Q. E. D.

(i would like to mention that i have not yet read chapter 10, so if i appeared to use anything from that chapter: it was a coincidence)